

UART TX Block

1. Test cases simulation waveform

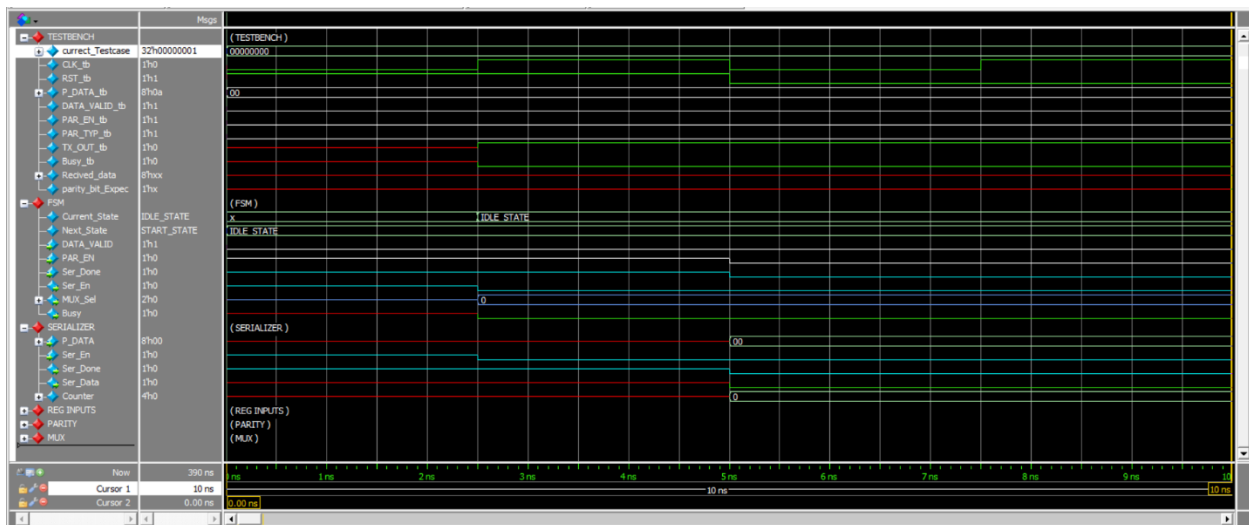


Figure 1 TESTCASE 0: Reset

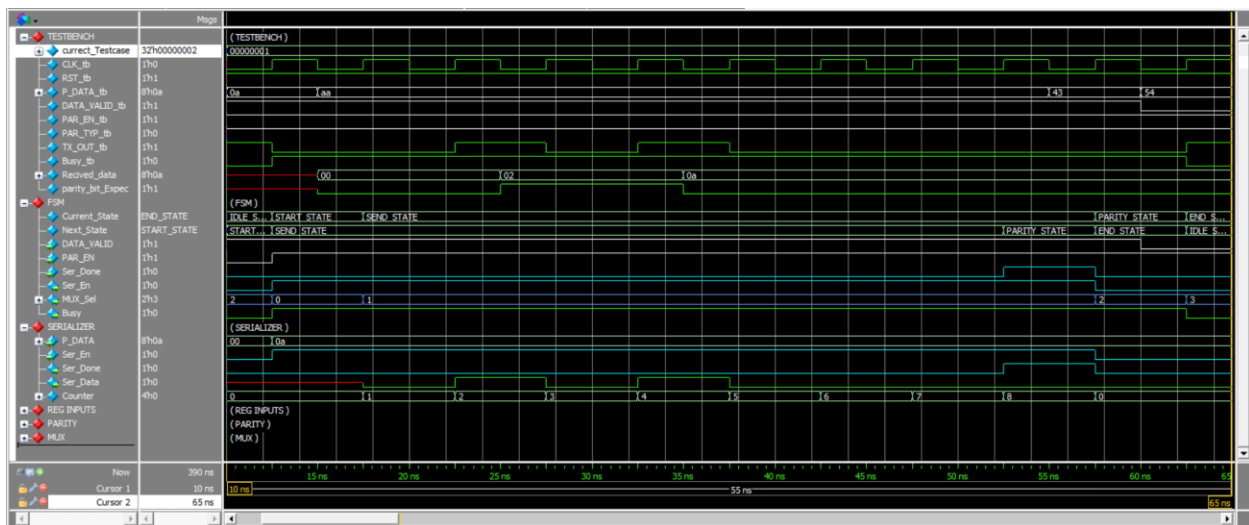


Figure 2 Testcase 1: expected to get an IDLE - START - SEND - PARITY - STOP

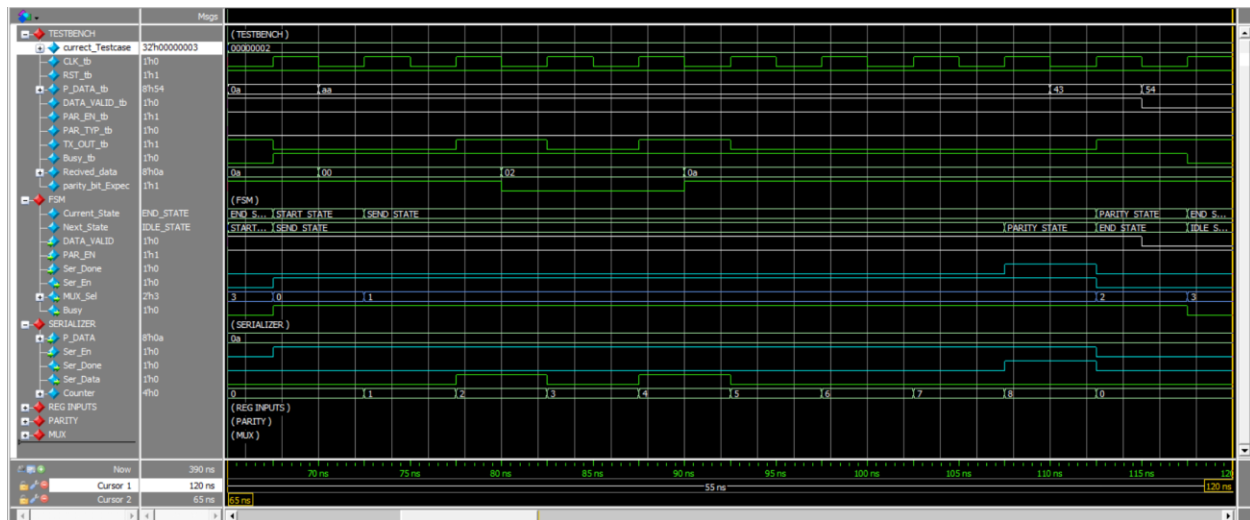


Figure 3 Testcase 2: expected to get an START - SEND - PARITY - STOP (IDLE skipped)

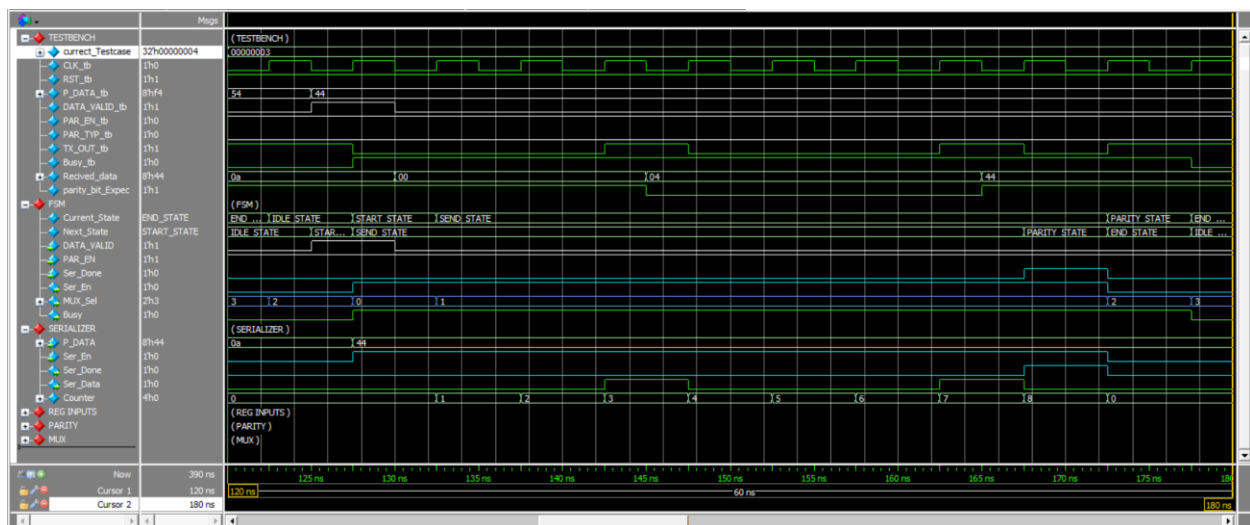


Figure 4 Testcase 3: expected to get an IDLE - START - SEND - PARITY - STOP (IDLE returned)

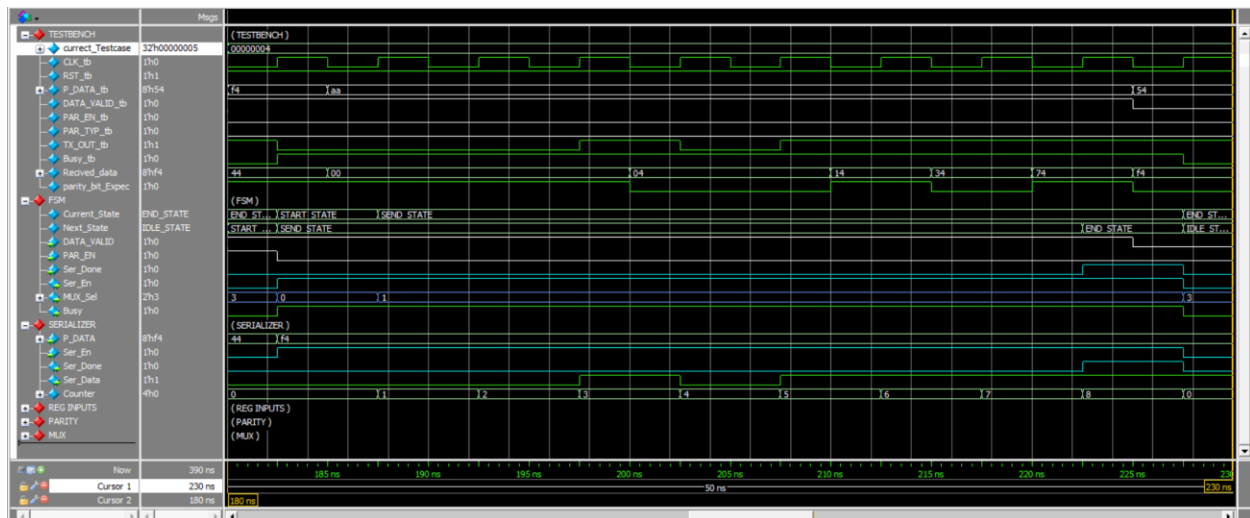


Figure 5 Testcase 4: expected to get an START - SEND – STOP (IDLE and PARITY skipped)

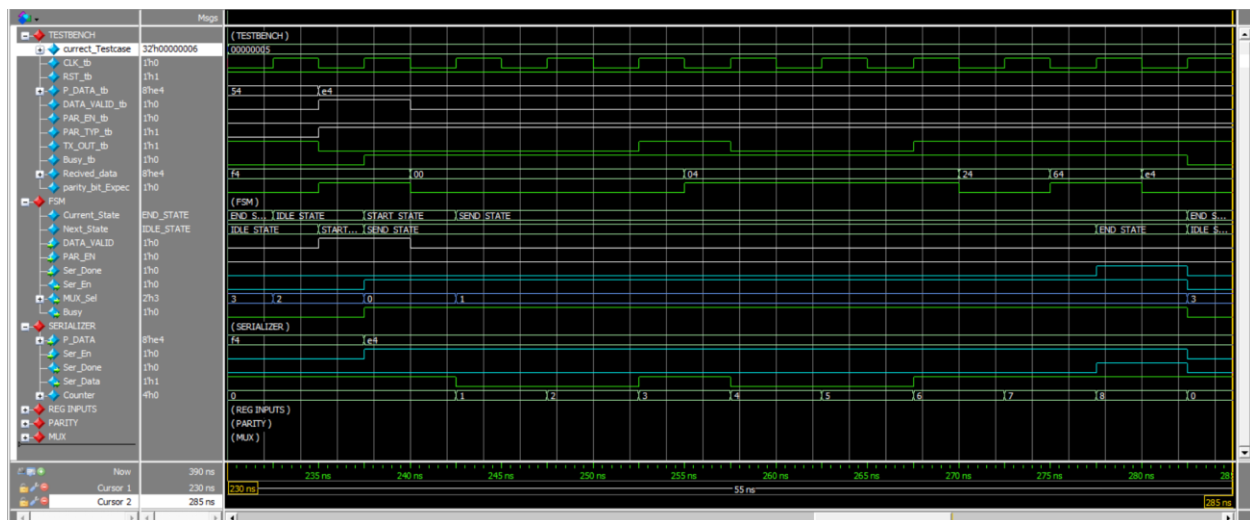


Figure 6 Testcase 5: expected to get an IDLE - START - SEND – STOP (PARITY skipped)

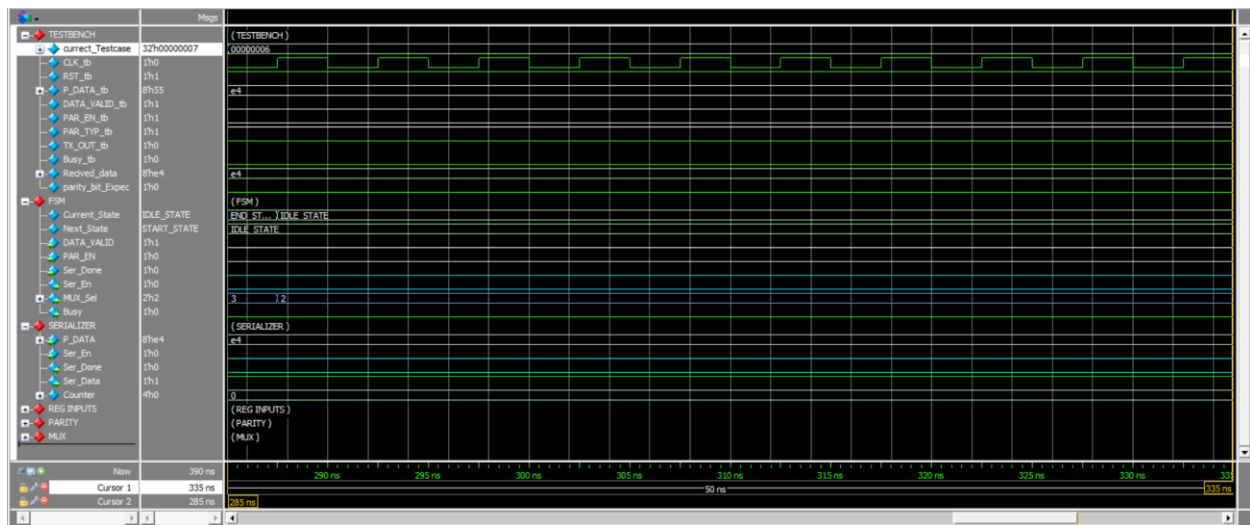


Figure 7 Testcase 6: expected to get an IDLE STATE and high TX_OUT and low Busy signal

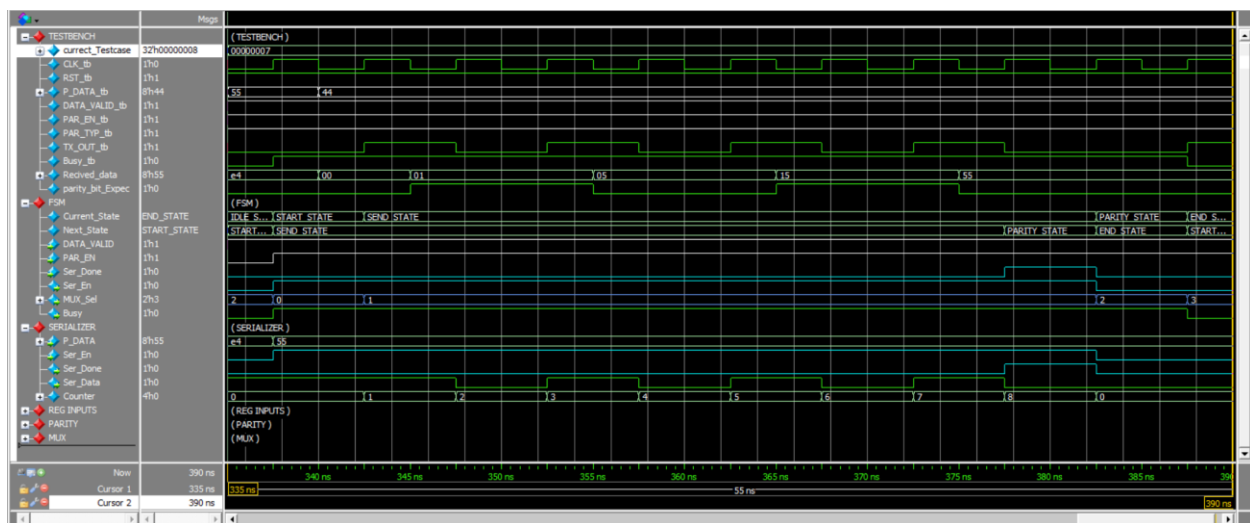


Figure 8 Testcase 7: expected to get an IDLE - START - SEND - PARITY - STOP (check the effect of always high DATA_VALID signal)

2. (Transcript) for Test cases

```
# Loading work.UART_TX_MTX(fast)
#
# Testcase 0: Reset succeeded
#
# Testcase 1 succeeded at Start bit: at time = 150 --- START_BIT = 0, Expected start bit = 0, DATA_VALID = 0, PAR_EN = 1, PAR_TYP = 1, RST = 1, Busy = 1
# Testcase 1 Succeeded at Recived data: at time = 550 --- inserted P_DATA = a, Recived data = a, current P_DATA = aa, DATA_VALID = 1, PAR_EN = 1, PAR_TYP = 1, RST = 1, Busy = 1
# Testcase 1 Succeeded at parity bit: at time = 600 --- parity bit = 0, Expected parity bit = 0, DATA_VALID = 1, PAR_EN = 1, PAR_TYP = 1, RST = 1, Busy = 1
# Testcase 1 Succeeded at Stop bit: at time = 650 --- STOP_BIT = 1, Expected stop bit = 1, DATA_VALID = 0, PAR_EN = 1, PAR_TYP = 1, RST = 1, Busy = 0
# Testcase 2 succeeded at Start bit: at time = 700 --- START_BIT = 0, Expected start bit = 0, DATA_VALID = 0, PAR_EN = 1, PAR_TYP = 0, RST = 1, Busy = 1
# Testcase 2 Succeeded at Recived data: at time = 1100 --- inserted P_DATA = a, Recived data = a, current P_DATA = aa, DATA_VALID = 1, PAR_EN = 1, PAR_TYP = 0, RST = 1, Busy = 1
# Testcase 2 Succeeded at parity bit: at time = 1150 --- parity bit = 1, Expected parity bit = 1, DATA_VALID = 1, PAR_EN = 1, PAR_TYP = 0, RST = 1, Busy = 1
# Testcase 2 Succeeded at Stop bit: at time = 1200 --- STOP_BIT = 1, Expected stop bit = 1, DATA_VALID = 0, PAR_EN = 1, PAR_TYP = 0, RST = 1, Busy = 0
# Testcase 3 succeeded at Start bit: at time = 1300 --- START_BIT = 0, Expected start bit = 0, DATA_VALID = 0, PAR_EN = 1, PAR_TYP = 0, RST = 1, Busy = 1
# Testcase 3 Succeeded at Recived data: at time = 1700 --- inserted P_DATA = 44, Recived data = 44, current P_DATA = 44, DATA_VALID = 0, PAR_EN = 1, PAR_TYP = 0, RST = 1, Busy = 1
# Testcase 3 Succeeded at parity bit: at time = 1750 --- parity bit = 1, Expected parity bit = 1, DATA_VALID = 0, PAR_EN = 1, PAR_TYP = 0, RST = 1, Busy = 1
# Testcase 3 Succeeded at Stop bit: at time = 1800 --- STOP_BIT = 1, Expected stop bit = 1, DATA_VALID = 0, PAR_EN = 1, PAR_TYP = 0, RST = 1, Busy = 0
# Testcase 4 succeeded at Start bit: at time = 1850 --- START_BIT = 0, Expected start bit = 0, DATA_VALID = 0, PAR_EN = 0, PAR_TYP = 0, RST = 1, Busy = 1
# Testcase 4 Succeeded at Recived data: at time = 2250 --- inserted P_DATA = f4, Recived data = f4, current P_DATA = aa, DATA_VALID = 1, PAR_EN = 0, PAR_TYP = 0, RST = 1, Busy = 1
# Testcase 4 Succeeded at Stop bit: at time = 2300 --- STOP_BIT = 1, Expected stop bit = 1, DATA_VALID = 0, PAR_EN = 0, PAR_TYP = 0, RST = 1, Busy = 0
#
# Testcase 5 succeeded at Start bit: at time = 2400 --- START_BIT = 0, Expected start bit = 0, DATA_VALID = 0, PAR_EN = 0, PAR_TYP = 1, RST = 1, Busy = 1
# Testcase 5 Succeeded at Recived data: at time = 2800 --- inserted P_DATA = e4, Recived data = e4, current P_DATA = e4, DATA_VALID = 0, PAR_EN = 0, PAR_TYP = 1, RST = 1, Busy = 1
# Testcase 5 Succeeded at Stop bit: at time = 2850 --- STOP_BIT = 1, Expected stop bit = 1, DATA_VALID = 0, PAR_EN = 0, PAR_TYP = 1, RST = 1, Busy = 0
# Testcase 6 succeeded at check state: at time = 3350 --- P_DATA = 11100100, DATA_VALID = 0, PAR_EN = 0, PAR_TYP = 1, RST = 1, TX_OUT = 1, Busy = 0
# Testcase 7 succeeded at Start bit: at time = 3400 --- START_BIT = 0, Expected start bit = 0, DATA_VALID = 1, PAR_EN = 1, PAR_TYP = 1, RST = 1, Busy = 1
# Testcase 7 Succeeded at Recived data: at time = 3800 --- inserted P_DATA = 55, Recived data = 55, current P_DATA = 44, DATA_VALID = 1, PAR_EN = 1, PAR_TYP = 1, RST = 1, Busy = 1
# Testcase 7 Succeeded at parity bit: at time = 3850 --- parity bit = 0, Expected parity bit = 0, DATA_VALID = 1, PAR_EN = 1, PAR_TYP = 1, RST = 1, Busy = 1
# Testcase 7 Succeeded at Stop bit: at time = 3900 --- STOP_BIT = 1, Expected stop bit = 1, DATA_VALID = 1, PAR_EN = 1, PAR_TYP = 1, RST = 1, Busy = 0
# ^ Note: dstop : UART_TX_tb.v(73)
# Time: 390 ns Iteration: 1 Instance: /UART_TX_tb
# Break in Module UART_TX_tb at UART_TX_tb.v line 73
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